Hybrid Variational Quantum-Classical Framework with Adaptive Weighting and Efficiency Assessment

Dilli Hang Rai
Institute of Science and Technology
Tribhuvan University, Kritipur, Nepal
dillihangrae@gmail.com

***Abstract*—Hybrid quantum-classical neural networks have emerged as a promising approach for leveraging quantum computing in machine learning while mitigating current hardware limitations. This paper presents Sim-HVQC, a hybrid Deep Quantum Neural Network that couples an adaptive, parameter-free SimAM weighting module with classical feature extraction to preserve class-discriminative information prior to encoding into a Variational Quantum Circuit (VQC). Previous studies are restricted to binary classification [1] [2] [3] [4] [5]. In contrast, the proposed framework is trained and evaluated on various multi-class datasets(MNIST, KMNIST, Fashion-MNIST, and EMNIST). The framework further demonstrates reproducibility, parameter efficiency, and interpretability through multi-seed evaluation, parameter analysis, and latent/quantum feature inspection. The source code is publicly available at** https://github.com/Dilli822/SimAM-HVQC.

***Index Terms*—SimAM, hybrid,VQC, quantum, multi-class**

I. Introduction

Quantum machine learning offers one such direction and hybrid quantum neuralnetworks, which combine classical neural-network components with quantum information processing units, have emerged as a practical framework for near-term quantum technologies [6]. Existing HQNN studies are commonly evaluated on binary-classification settings or a single benchmark dataset, while reproducibility analysis, efficiency assessment, and model interpretability remain comparatively underexplored in hybrid quantum-classical image-classification frameworks. In particular, [5] focused on binary classification, while [7] performed multi-class classification using a single dataset. Both studies reported only training accuracy, without validation or test results. Although [8] achieved impressive accuracy, its evaluation was limited to binary classification. Similarly, [9] conducted experiments across datasets but remained restricted to binary classification. These previous studies highlight the need for dataset-level experiments involving a larger number of multi-class, multi-dataset evaluations with empirical validation and test performance.

The study [10] demonstrated that different hybrid quantum-classical architectures exhibit varying performance characteristics under different circuit configurations. Therefore, continued exploration of improved hybrid quantum-classical architectures remains necessary. Motivated by the growing interest in practical hybrid quantum neural networks and the need to explore improved hybrid architectures [6], This paper proposes SimAM-HVQC, a parameter-efficient hybrid quantum-classical model for multi-class image classification that adopts an alternative approach by incorporating a parameter-free attention mechanism prior to quantum encoding.

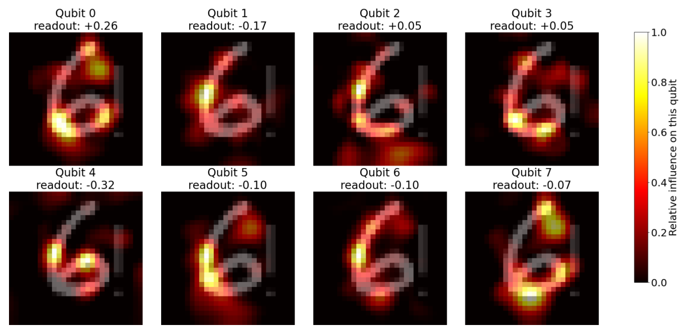

Fig. 1. Occlusion-based qubit sensitivity maps for a sample input. The varying sensitivity patterns across qubits indicate that different quantum measurements capture complementary information from the compressed latent representation

The contributions of this paper are listed below:

- Proposed a hybrid Deep Quantum Neural Network (DQNN) that integrates a SimAM attention-enhanced classical feature extractor with a Variational Quantum Circuit (VQC).
- Evaluated the framework on multiple standard benchmarks with varying numbers of classes:MNIST,Fashion-MNIST(F-MNIST), KMNIST and EMNIST.
- Assessed model reproducibility through multi-seed experiments, analyzed performance under varying training-data sizes, and provided a lightweight interpretability analysis of latent feature representations and quantum measurement outputs.

II. Methodology

This section describes the dataset preparation, classical feature extraction, and the proposed hybrid quantum-classical architecture, along with the corresponding algorithm.

A. Dataset

The proposed framework was evaluated on MNIST, Fashion-MNIST, KMNIST and EMNIST using a standard 70:15:15 train-validation-test split. MNIST, Fashion-MNIST, and KMNIST are 10-class, 28×28 grayscale datasets; EMNIST Balanced has 47 classes of 28×28 grayscale character images.

B. SimAM Attention

Before dimensionality reduction, the input representation is refined using the parameter-free Simple Attention Module (SimAM). Given an input feature map

$$X \in \mathbb{R}^{C \times H \times W}, \tag{1}$$

SimAM computes neuron-wise attention coefficients using an energy-based formulation [11]. The channel-wise mean and variance are

$$\mu = \frac{1}{M}\sum_{i=1}^{M} x_i, \qquad \sigma^2 = \frac{1}{M}\sum_{i=1}^{M}(x_i - \mu)^2, \tag{2}$$

where $M = H \times W$. For neuron x_i, the inverse energy is

$$E_i^{-1} = \frac{(x_i - \mu)^2}{4(\sigma^2 + \lambda)} + \frac{1}{2}, \tag{3}$$

where λ is a regularization parameter. The attention coefficient is obtained as

$$A_i = \sigma(E_i^{-1}), \tag{4}$$

where $\sigma(\cdot)$ denotes the sigmoid function. The attention-refined representation is then computed by

$$X_{\mathrm{SimAM}} = X \odot A, \tag{5}$$

where $\odot$ denotes element-wise multiplication. Consequently, activations exhibiting greater distinguishability from the underlying channel distribution receive larger attention coefficients. The resulting representation X_{SimAM} is forwarded to the classical feature extractor for subsequent compression and quantum encoding.

C. Classical Neural Network

The classical neural network $f_{\mathrm{feat}}(\cdot)$ functions as a learnable feature extractor and representation learner, transforming the SimAM-enhanced input $\mathbf{X}_a$ into a discriminative Q-dimensional latent embedding $\mathbf{F} = f_{\mathrm{feat}}(\mathbf{X}_a) \in \mathbb{R}^Q$ that supports downstream quantum processing and classification.

D. Quantum Processing

The compressed latent representation $F \in \mathbb{R}^Q$ is encoded into a variational quantum circuit (VQC), enabling feature transformation in a high-dimensional Hilbert space. The encoded quantum state evolves as

$$|\psi\rangle = U(\omega, \theta)|0\rangle^{\otimes Q}, \tag{6}$$

where $\theta = \pi \tanh(F)$ and $U(\cdot)$ denotes the trainable quantum circuit. The VQC exploits superposition and entanglement to capture complex feature interactions, producing quantum-enhanced representations that are difficult to express through the preceding low-dimensional classical mapping alone. Quantum features are extracted through Pauli-Z expectation measurements,

$$z_i = \langle \psi | Z_i | \psi \rangle, \qquad i = 1, \ldots, Q, \tag{7}$$

and forwarded to the final classifier. In the proposed framework, the quantum module employs 8 qubits and 6 strongly entangling layers, resulting in 144 trainable quantum parameters.

Algorithm 1 Proposed SIMAM-HQNN with a VQC

Require: Input image X; number of qubits Q; number of quantum layers L; number of classes C
Ensure: Predicted class scores $\hat{Y}$
$X_a \leftarrow \mathrm{SIMAM}(X)$
$\mathbf{x} \leftarrow \mathrm{Flatten}(X_a)$
$\mathbf{F} \leftarrow \mathrm{FC}_{784\to256\to128\to Q}(\mathbf{x})$
$\boldsymbol{\theta} \leftarrow \pi \tanh(\mathbf{F})$
Encode $\boldsymbol{\theta}$ on Q qubits using angle embedding
Apply L strongly entangling layers
$\mathbf{Z} \leftarrow \{\langle Z_1 \rangle, \ldots, \langle Z_Q \rangle\}$
$\hat{Y} \leftarrow \mathrm{FC}_{Q\to64\to C}(\mathbf{Z})$
return $\hat{Y}$

TABLE I
MODEL CONFIGURATION, DATASET, AND HYPERPARAMETER SUMMARY

Parameter	Value
Dataset	
Dataset used	MNIST, KMNIST, EMNIST, FashionMNIST
Input image dimension	28×28 (grayscale)
Flattened input size	784
Number of classes	10, 47
Layer Dimensions (In $\to$ Out)	
Classical encoder L0	$784 \to 256$
Classical encoder L1	$256 \to 128$
Classical encoder L2	$128 \to 8$
Quantum layer	$8 \to 8$ (qubits)
Classifier L0	$8 \to 64$
Classifier L1	$64 \to$ number of dataset classes
Training Hyperparameters	
Random seed	42, 32, 102, 72, 2
Batch size	64
Epochs	25
Learning rate	0.001
Optimizer	Adam
Loss Function	Weighted Cross-Entropy Loss
Device	CPU
Train/Validation/Test Split	70:15:15
Gradient Clipping	1.0
Activation Function	ReLU
SIMAM Parameter	$\lambda = 10^{-4}$
Quantum Configuration	
Quantum Encoding	AngleEmbedding
Variational Ansatz	Strongly Entangling Layers
Measurement Observable	Pauli-Z Expectation Values
Differentiation Method	Backpropagation
Quantum Simulator	PennyLane `default.qubit`

III. EXPERIMENTAL SETUP

This section presents the experimental configuration adopted in the paper. Table I summarizes the datasets, model architecture, quantum-circuit configuration, and training hyperparameters.

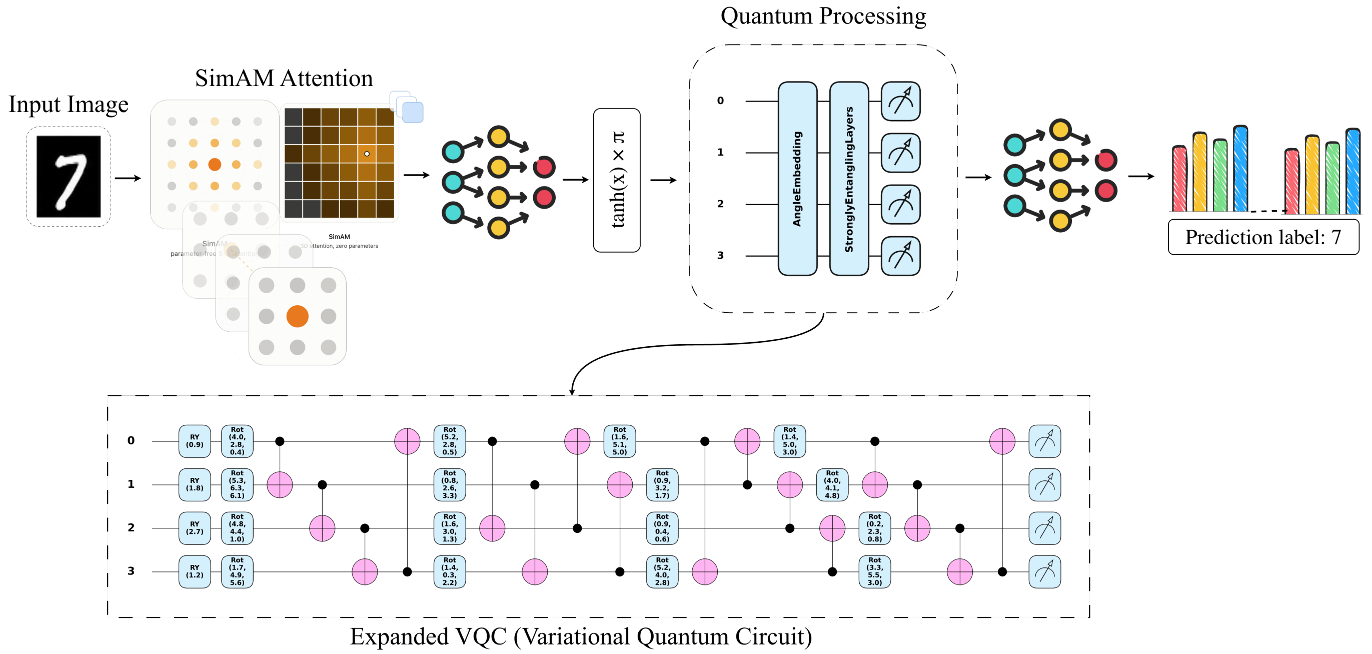

Fig. 2. Architecture of the proposed SimAM-enhanced hybrid variational quantum-classical neural network (SIMAM-HVQC). The input image is first processed by the parameter-free SIMAM attention module to enhance informative feature responses. The refined feature map is flattened and passed through a classical fully connected feature extractor, reducing the original 784-dimensional representation to 256, 128, and finally Q quantum-compatible features. The resulting features are scaled using $\boldsymbol{\theta} = \pi \tanh(\mathbf{F})$ and encoded into the quantum register through R_Y-based AngleEmbedding. The variational quantum component consists of Q qubits and L trainable Strongly Entangling Layers, followed by Pauli-Z expectation-value measurements to produce Q quantum features. These measurements are subsequently processed by a classical fully connected classifier $Q \to H \to C$, where H denotes the hidden-layer dimension and C the number of target classes, producing the final class prediction $\hat{Y}$. The lower panel provides the corresponding gate-level representation of the variational quantum circuit, illustrating the trainable single-qubit rotations and entangling operations.

IV. Results

This section presents the results obtained from the proposed method.

TABLE II
Comparison of Proposed Method Performance with the Existing Methods Performance

Study	Method	Qubits	Dataset	Classes	Loss	Acc. (%)
[1]	MPS-VQC	4	MNIST	2	0.3183	99.44
[12]	QCNN (Hybrid VQC)	12 qubits	MNIST	4	-	85.14
			MNIST	4	-	90.03
			F-MNIST	4	-	85.93
[13]	HQNN	16	MNIST	10	-	89.06
[8]	HQNN	5	MNIST	2	0.0274	99.21
[14]	SEQNN	-	MNIST	10	-	87.56
[15]	Hybrid QCNN	8	MNIST	4	-	88.52
		8	F-MNIST	4	-	86.55
		8	MNIST	4	-	93.55
Proposed	**SimAM-HVQC**	**8**	**MNIST**	**10**	**0.1437 ± 0.013**	**97.57 ± 0.05**
		8	**F-MNIST**	**10**	**0.3349± 0.0131**	**87.98 ± 0.08**
		8	**EMNIST**	**47**	**0.7212 ± 0.0236**	**80.16 ± 0.17**
		8	**KMNIST**	**10**	**0.3545 ± 0.0273**	**88.20 ± 0.61**

TABLE III
Performance at Different Dataset Utilization Levels

Dataset Utilized	KMNIST		F-MNIST		MNIST		EMNIST	
	Acc.(%)	Loss	Acc.(%)	Loss	Acc.(%)	Loss	Acc.(%)	Loss
10%	**75.09**	0.5083	83.07	0.5838	**94.46**	**0.2874**	83.66	**0.6001**
20%	**80.54**	0.4943	83.68	**0.5730**	**95.00**	**0.2769**	85.12	0.5421
30%	**83.72**	0.3934	85.28	**0.4551**	**96.10**	**0.2590**	85.44	0.4420
40%	**85.50**	0.4347	86.28	**0.4768**	**96.12**	**0.2382**	86.86	0.4530
50%	**86.47**	**0.4356**	86.75	0.3926	**96.35**	**0.2347**	86.47	0.3949
60%	87.87	**0.4344**	**87.08**	0.4261	**96.70**	**0.2730**	87.25	0.4231
70%	**86.47**	0.3414	87.24	**0.4127**	**96.88**	**0.1931**	**86.47**	0.33949
80%	**87.54**	0.3272	**87.54**	**0.4097**	**97.01**	**0.2128**	87.55	0.3898
90%	**87.18**	**0.3225**	**87.18**	**0.3927**	**97.09**	0.2012	87.31	0.3924

V. Reproducibility, Data-Experiment, Ablation Study and Representation Analysis

The dataset utilization analysis in Table III indicates that the proposed SimAM-HVQC exhibits robust learning characteristics, achieving reasonable performance with limited training data while benefiting from increased dataset availability.

The ablation results of Table V verify that the inclusion of the parameter-free SimAM attention mechanism contributes positively to model effectiveness, leading to improved accuracy and reduced classification loss across multiple benchmark datasets.

Fig. 3 shows the evolution of latent representations across SimAM-HVQC. The input space exhibits substantial class overlap (16.8% PCA variance), whereas intermediate and final representations become increasingly structured. Notably, the 8-dimensional quantum representations retain high projected variance (52.0–62.9%) and clear class organization despite substantial dimensionality reduction. The classifier latent space exhibits the most distinct class clusters.

Occlusion-based qubit sensitivity analysis illustrates the image regions (Fig. 1) that most strongly influence individual quantum measurements. Distinct qubits exhibit sensitivity to different portions of the input pattern, suggesting that the variational quantum circuit captures complementary feature representations from the compressed latent embedding. The

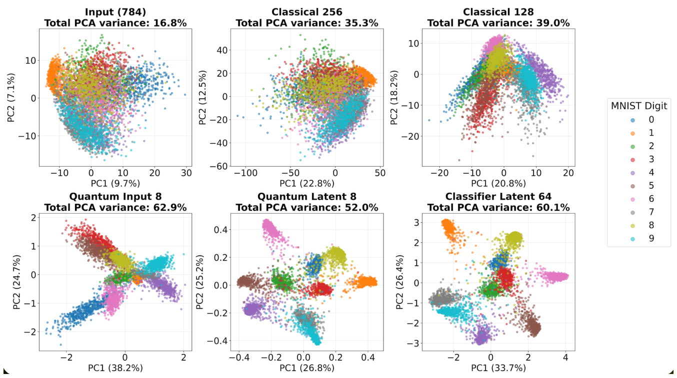

Fig. 3. PCA of latent representations across the hybrid quantum-classical network. Each panel projects a layer's feature space onto its top two principal components (variance explained on axes), colored by digit label. Class separability increases with depth, with the quantum layer and final classifier layer achieving the clearest separation despite low dimensionality.

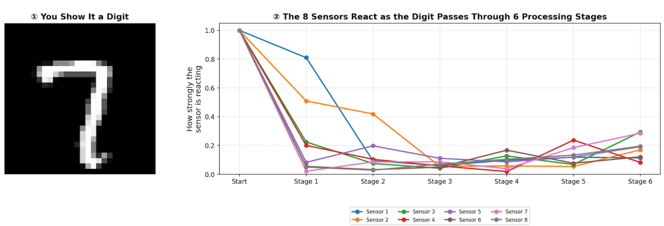

Fig. 4. Shows that the eight quantum measurements progressively develop distinct response patterns across the six variational layers, indicating specialization of quantum features and formation of a discriminative quantum representation for classification.

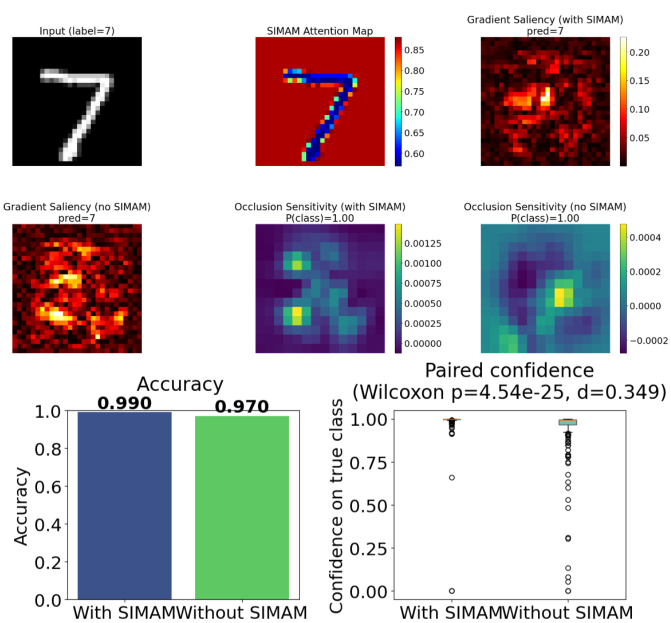

Fig. 5. Top Figure: Attention, saliency, and occlusion maps illustrate feature importance for a sample input. Bottom Figure: The dataset-level comparison for 'n=200' sample test-dataset shows improved accuracy and significantly higher prediction confidence when SimAM is enabled (Wilcoxon $p = 4.54 \times 10^{-25}$, Cohen's $d = 0.349$).

TABLE IV
EFFICIENCY ASSESSMENT OF THE PROPOSED SIM-HVQC ON CPU

Metric	**SIM-HVQC**
Parameters	236,258
Qubits	8
Quantum Layers	6
Quantum Parameters	144
Test Samples	2,000
Inference Time (s)	1.6249
Latency (ms/sample)	0.8124
Throughput (samples/s)	1230.85

TABLE V
ABLATION STUDY OF SIM-HVQC WITH AND WITHOUT SIMAM

Dataset	SIM-HVQC		No SIM-HVQC	
	Accuracy (%)	Loss	Accuracy (%)	Loss
MNIST	**97.57 ± 0.05**	**0.1437 ± 0.0139**	97.48 ± 0.18	0.1638 ± 0.0054
Fashion-MNIST	**87.98 ± 0.08**	**0.3345 ± 0.0132**	87.51 ± 0.11	0.4424 ± 0.0218
EMNIST	**80.16 ± 0.17**	**0.7212 ± 0.0236**	78.99 ± 0.06	0.8921 ± 0.0302
KMNIST	88.20 ± 0.63	**0.3544 ± 0.0283**	**88.23 ± 0.57**	0.9284 ± 0.0593

test result shows of the bottom panel Figure 5 that enabling SimAM produces a statistically significant improvement in prediction confidence (and related performance) compared to the model without SimAM. SimAM-HVQC maintains computational efficiency (Table IV) despite incorporating quantum processing. The low quantum parameter count together with sub-millisecond inference latency suggests that the quantum component introduces limited computational overhead.

Evolution of quantum feature representations across the variational circuit. Fig. 4 shows the layer-wise response magnitude of the eight qubits, revealing distinct feature-specialization patterns. The lower-left panel of Fig. 4 illustrates the growth of inter-qubit correlations throughout the circuit depth, while the lower-right panel presents the final Pauli-Z measurement outputs that form the quantum feature vector supplied to the classifier.

VI. CONCLUSION

In conclusion, the proposed SimAM-HVQC demonstrated competitive and promising performance across multiple standard benchmark datasets while maintaining parameter efficiency and supporting reproducibility. Efficiency and interpretability analyses highlights its potential as a practical hybrid quantum-classical framework for representation learning and visual recognition tasks.

REFERENCES

[1] S. Y.-C. Chen, C.-M. Huang, C.-W. Hsing, and Y.-J. Kao, "Hybrid quantum-classical classifier based on tensor network and variational quantum circuit," 2020. [Online]. Available: https://arxiv.org/abs/2011.14651

[2] C.-W. H. Samuel Yen-Chi Chen, Chih-Min Huang and Y.-J. Kao, "An end-to-end trainable hybrid classical-quantum classifier," *Machine Learning: Science and Technology*, vol. 2, no. 4, p. 045021, sep 2021. [Online]. Available: https://doi.org/10.1088/2632-2153/ac104d

[3] D. Ranga, S. Prajapat, Z. Akhtar, P. Kumar, and A. V. Vasilakos, "Hybrid quantum–classical neural networks for efficient mnist binary image classification," *Mathematics*, vol. 12, no. 23, 2024. [Online]. Available: https://www.mdpi.com/2227-7390/12/23/3684

[4] C. Long, M. Huang, X. Ye, Y. Futamura, and T. Sakurai, "Hybrid quantum-classical-quantum convolutional neural networks," *Scientific Reports*, vol. 15, no. 1, p. 31780, 2025. [Online]. Available: https://doi.org/10.1038/s41598-025-13417-1

[5] M. Das and T. Bolisetti, "Variational quantum neural networks (vqnns) in image classification," 2023. [Online]. Available: https://arxiv.org/abs/2303.05860

[6] L. Monbroussou, M. Periyasamy, V. Kuzmin, P. Sekatski, V. Patapovich, A. Sagingalieva, and A. Melnikov, "Hybrid quantum neural networks: Theory, implementations, and applications," 2026. [Online]. Available: https://arxiv.org/abs/2608.01194

[7] S. Choe, "Continuous variable quantum mnist classifiers," 2022. [Online]. Available: https://arxiv.org/abs/2204.01194

[8] A. Senokosov, A. Sedykh, A. Sagingalieva, B. Kyriacou, and A. Melnikov, "Quantum machine learning for image classification," *Machine Learning: Science and Technology*, vol. 5, no. 1, p. 015040, Mar. 2024. [Online]. Available: http://dx.doi.org/10.1088/2632-2153/ad2aef

[9] M. A. Hafeez, A. Munir, and H. Ullah, "H-qnn: A hybrid quantum–classical neural network for improved binary image classification," *AI*, vol. 5, no. 3, pp. 1462–1481, 2024. [Online]. Available: https://www.mdpi.com/2673-2688/5/3/70

[10] K. Zaman, T. Ahmed, M. A. Hanif, A. Marchisio, and M. Shafique, "A comparative analysis of hybrid-quantum classical neural networks," 2024. [Online]. Available: https://arxiv.org/abs/2402.10540

[11] L. Yang, R.-Y. Zhang, L. Li, and X. Xie, "Simam: A simple, parameter-free attention module for convolutional neural networks," vol. 139, pp. 11 863–11 874, 18–24 Jul 2021. [Online]. Available: https://proceedings.mlr.press/v139/yang21o.html

[12] D. Bokhan, A. S. Mastiukova, A. S. Boev, D. N. Trubnikov, and A. K. Fedorov, "Multiclass classification using quantum convolutional neural networks with hybrid quantum-classical learning," *Frontiers in Physics*, vol. Volume 10 - 2022, 2022.

[13] Y. Zeng, H. Wang, J. He, Q. Huang, and S. Chang, "A multi-classification hybrid quantum neural network using an all-qubit multi-observable measurement strategy," *Entropy*, vol. 24, no. 3, p. 394, 2022.

[14] Q. Bai and X. Hu, "Superposition-enhanced quantum neural network for multi-class image classification," *Chinese Journal of Physics*, vol. 89, pp. 378–389, 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0577907324001151

[15] S. Anwar, S. Das, M. I. Hossain, and J. Mahmud, "Hybrid quantum-classical learning for multiclass image classification," 8 2025. [Online]. Available: https://arxiv.org/abs/2508.18161